

Operational Criteria and Experimental Benchmarks for Quantifying Self-Learning in Hierarchical Spintronic Ensembles (SHOA-Validate)

Konstantin Fedotov (ORCID: 0009-0001-6376-7968)

Abstract

The transition from theoretical models of self-healing to their practical implementation requires stringent, reproducible verification criteria. This work closes this gap for the SHOA-Neuro protocol. We propose a set of four operational metrics: (1) Recovery Fidelity threshold > 0.9 ; (2) integral Neuro-Stability Index S_{neuro} ; (3) formalized memory law $K_{\text{eff}}(N)$ with exponential saturation; and (4) coherent activation criterion $M_{\text{spin}} > M_{\text{crit}}$. For each metric, the physical meaning, mathematical definition, and specific experimental endpoints suitable for verification on femtosecond pump-probe setups are defined.

1. Introduction

Publications on the SHOA (Self-Healing Optimal Adaptive) project have shown that the adaptive reduction of the recovery threshold is a direct analogue of long-term potentiation (LTP) in neurons [1, 2]. However, reviewers regularly ask two questions:

1. "Is this not just a successful parameter fit?"
2. "How to distinguish true learning from random noise?"

The answers to these questions lie not in additional mathematical formalism, but in reproducible experimental criteria. SHOA-Validate provides such criteria.